

Program No: 4

File Name:

Ex. No:

Date: _____

DEVELOP A DYNAMIC SPA (SINGLE PAGE APPLICATION) USING REACT.JS.

AIM:

The aim of developing a dynamic SPA using React.js is to build a fast, interactive web app that loads a single HTML page and updates content dynamically without full page reloads, providing a smooth user experience and better performance.

ALGORITHM:

Step 1: Set up the React development environment with Node.js and Create React App.

Step 2: Structure the app with reusable components like Header, Footer, and Main content.

Step 3: Use React Router for client-side navigation without page reloads.

Step 4: Create individual component pages for dynamic content display.

Step 5: Manage app state efficiently using React's state or external libraries.

Step 6: Add styling and optimize performance with minified CSS/JS.

Step 7: Test the SPA on modern browsers and deploy it to a hosting platform.

PROGRAM:

src/main.jsx

```
import React from "react";
```

```
import ReactDOM from "react-dom/client";
```

```
import { BrowserRouter } from "react-router-dom";
```

```
import App from "./App";
```

```
import "./index.css";
```

```
ReactDOM.createRoot(document.getElementById("root")).render(
```

```
  <BrowserRouter>
```

```
    <App />
```

```
  </BrowserRouter>
```

```
);
```

src/App.jsx

```
import { Routes, Route } from "react-router-dom";
```

```
import Navbar from "./components/Navbar";
```

```
import Footer from "./components/Footer";
```

```
import Home from "./pages/Home";
```

```
import About from "./pages/About";
```

```
import Services from "./pages/Services";
```

```
import Contact from "./pages/Contact";
```

```
export default function App() {
```

```
  return (
```

```
    <div className="flex flex-col min-h-screen bg-gray-50">
```

```

<Navbar />

<main className="flex-1 p-6 container mx-auto">

  <Routes>

    <Route path="/" element={<Home />} />

    <Route path="/about" element={<About />} />

    <Route path="/services" element={<Services />} />

    <Route path="/contact" element={<Contact />} />

  </Routes>

</main>

<Footer />

</div>

);

}

src/pages/Home.jsx

import data from "../data";

import Card from "../components/Card";

export default function Home() {

  return (

    <div>

      <h2 className="text-3xl font-bold mb-6 text-center text-blue-700">

        Welcome to Dynamic SPA

      </h2>

      <div className="grid gap-6 md:grid-cols-3 sm:grid-cols-2">

        {data.map((item) => (

```

```
<Card key={item.id} title={item.title} description={item.description} />
```

```
    )}
```

```
</div>
```

```
</div>
```

```
);
```

```
}
```

src/components/Navbar.jsx

```
import { Link, useLocation } from "react-router-dom";
```

```
export default function Navbar() {
```

```
  const location = useLocation();
```

```
  const linkClass = (path) =>
```

```
    `px-3 py-2 rounded hover:bg-blue-600 ${
```

```
      location.pathname === path ? "bg-blue-700" : ""
```

```
    }`;
```

```
  return (
```

```
    <nav className="bg-gray-900 text-white flex justify-between items-center p-4 shadow-md">
```

```
      <h1 className="text-xl font-bold">Dynamic SPA</h1>
```

```
      <div className="space-x-2">
```

```
        <Link to="/" className={linkClass("/")}>
```

```
          Home
```

```
        </Link>
```

```
<Link to="/about" className={linkClass("/about")}>
```

```
  About
```

```
</Link>
```

```
<Link to="/services" className={linkClass("/services")}>
```

```
  Services
```

```
</Link>
```

```
<Link to="/contact" className={linkClass("/contact")}>
```

```
  Contact
```

```
</Link>
```

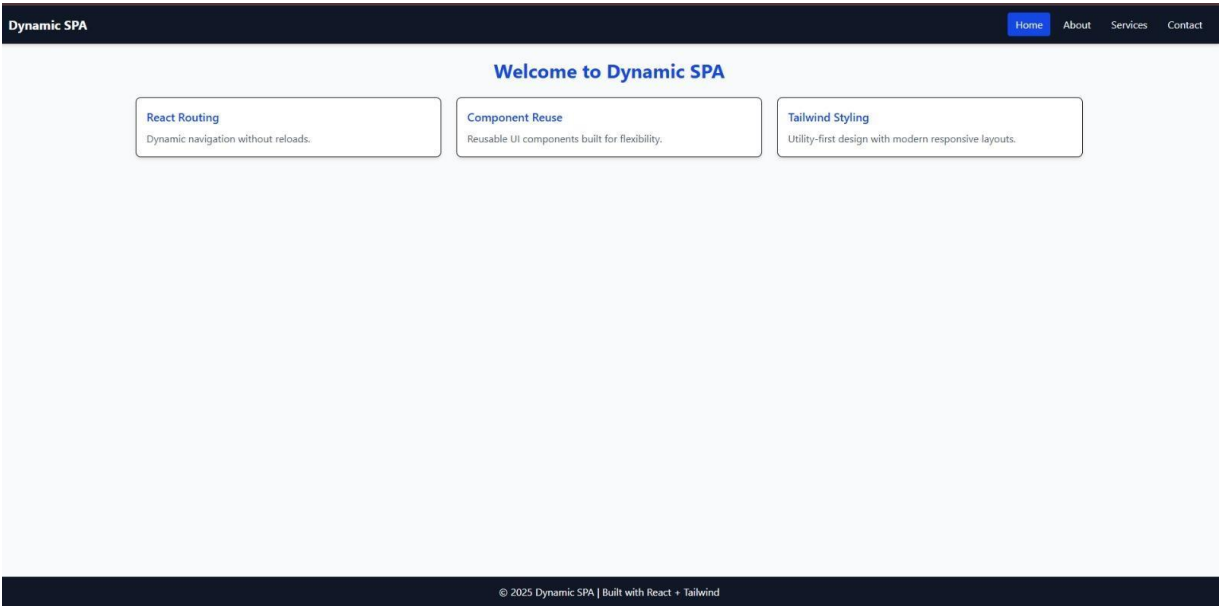
```
</div>
```

```
</nav>
```

```
);
```

```
}
```

OUTPUT:



RESULT:

Thus the program was successfully executed